Clinton Enwerem

enwerem@umd.edu | 301-416-9430 | clintonenwerem.com | GitHub: @coenwerem | LinkedIn | Scholar

Summary

Robotics Research Engineer with 5+ years of research experience and Electrical and Computer Engineering Ph.D. candidate at the University of Maryland, graduating in Fall 2026. Develop demonstration data pipelines, evaluate imitation and reinforcement learning policies, and implement Python/C++ sensing and control software for dexterous manipulation in simulation and on physical, poly-articulated robots.

Skills

Robotics: ROS/ROS 2, ros2_control, MoveIt 2, MoveIt Servo, SocketCAN, VR teleoperation; **Planning & Control:** RRT*, FCL, state estimation, CBF-CLF QPs, MPC; **Learning:** PyTorch, ACT, DAgger, Diffusion Policy, PPO, flow matching; **Simulation:** Isaac Lab/Sim, MuJoCo, Drake; **Software:** Python, C++, Linux, Docker, git, CI; **Robots:** Fairino FR3, xArm7, UR3e, UR5, Robotiq Hand-E, RealHand A7 (Bimanual Arm-Hand System), RealHand L6, LEAP Hand v1, Unitree G1 (simulation).

Work Experience

Graduate Research Assistant | Aug. 2021 – Present
Institute for Systems Research (ISR), University of Maryland | College Park, MD

• Develop grasp-synthesis, state-estimation, motion-planning, and feedback-control methods for dexterous manipulation under contact uncertainty; implement Python/C++ and ROS 2 software for physical arm–hand systems.
• Trained PPO with GDF rewards in **1,024 Isaac Lab environments**: **64/64** pregrasp arrivals versus **0/64** with palm-distance rewards. Published distributional RL (**CDC 2025**): **448 goals** versus **329/239** across **100 MuJoCo episodes**, with similar constraint-violation costs.
• Developed VR teleoperation software for an xArm7-RealHand-L6 robotic manipulator with wrist-to-Cartesian-velocity mapping, relative-joint-position hand control, grasp primitives, thumb-abduction control, and hand-state reset; integrated MoveIt Servo with Dimensional OS (DimOS) for real-time collision detection, arm speed modulation, and velocity damping near kinematic boundaries.

Research Intern | Jun. – Aug. 2022
MATRIX Lab, USM at Southern Maryland | California, MD

• Identified a discrete-time Crazyflie model from flight data and implemented a finite-horizon optimal formation controller in Python and Crazyswarm; executed leader–follower formation flights with 10 physical Crazyflie quadrotors under localization uncertainty.

Robotics Engineer | Mar. 2020 – Feb. 2021
Kognitive Robotics | Lagos, Nigeria

• Designed power and sensor–actuator interfaces in Altium, assembled a mobile robot, and implemented and tested its ROS software on physical hardware.

Robotics Trainee | Jan. 2020 – Feb. 2021
Robotics & Artificial Intelligence Nigeria (RAIN) | Ibadan, Nigeria

• Built ROS-based visual-SLAM, state-estimation, and flight-control software for a quadrotor's onboard computer.

Projects

ParcelStow — Temporal Robustness in Imitation Learning (Paper, project page, Code, Data & Models) | *Isaac Lab, PyTorch* 2026
• Authored ParcelStow, an Isaac Lab benchmark analyzing temporal robustness in imitation learning; generated **970K** expert states (**297** demonstrations) and trained three imitation baselines. At 2× execution speed, ACT success fell to **53%** versus **84%** for the expert, with failures concentrated in high-precision insertion.

Grasp Distance Fields — Feedback Grasp Execution (Project, Paper) | *C++, Python, QP* 2026
• Implemented a CBF-CLF quadratic-program controller for feedback grasp execution; simulated experiments lifted **46 of 50** objects, with a median QP solve time of **0.09 ms**.

realhand_l6_ros2 — Dexterous-Hand ROS 2 Software (Code) | *C++, SocketCAN* 2026
• Implemented joint/tactile-state interfaces, contact-gated control, and MoveIt 2 configuration with mock and virtual-CAN CI tests. Verified communication, feedback, tactile acquisition, and trajectory execution on a physical RealHand L6.

Selected Publications

• **C. Enwerem et al.** "Does Imitation Learning Preserve Temporal Robustness in Dexterous Manipulation? An Expert–Learner Comparison Across Task Execution Speeds." *arXiv preprint*, 2026. arXiv:2609.01453, project page, code, data & models.
• **C. Enwerem et al.** "Grasp Execution Without a Planner: Configuration-Space Grasp Distance Fields with Certified Safety & Guaranteed Quality." *Preprint*, 2026. arXiv:2608.00600, project page.
• **C. Enwerem et al.** "Safety-Aware Reinforcement Learning for Control via Risk-Sensitive Action-Value Iteration and Quantile Regression." *CDC 2025*. arXiv:2506.06954.
• Additional first-author publications covering control, safe learning, and robotic manipulation: IROS 2026, CDC 2026, CDC 2025, CDC 2024; full publication list on Google Scholar.

Education

Ph.D., Electrical & Computer Engineering, University of Maryland, College Park, MD. Expected Fall 2026. Aug. 2021 – Present
B.Eng., Electrical Engineering, University of Nigeria, Nsukka. Highest Honors. GPA 3.84/4.0. Aug. 2018

Clinton Enwerem
4448 A. V. Williams Bldg., 8223 Paint Branch Dr., College Park, MD 20742
enwerem@umd.edu | 301-416-9430 (Mobile)

Dear Technical Recruiter,

Application: Research Scientist, Robotics

I work on dexterous manipulation, robot learning, motion planning, state estimation, and feedback control. I am an Electrical and Computer Engineering Ph.D. candidate at the University of Maryland, advised by Dr. John Baras and Dr. Calin Belta and graduating in Fall 2026. I formulate technical questions, implement the resulting methods in Python and C++, and evaluate them in simulation and on physical arms with dexterous hands.

In ParcelStow, I built an Isaac Lab Unitree G1 and RealHand L6 grasp–reorient–insert task, collected 297 successful expert demonstrations, and trained DAgger, Diffusion Policy, and ACT. The expert and one ACT policy both reached 100% nominal success. At twice nominal speed, expert success was 84% and ACT success was 53%, a 31-percentage-point difference within the demonstrated speed range. Stage outcomes placed 35 of ACT's 47 failures during insertion.

My manipulation research also addresses uncertain contact and feedback execution. FIRMGrasp uses a CVaR wrench-space margin to rank grasps under friction uncertainty. Grasp Distance Fields use a CBF-CLF quadratic program during execution; simulated experiments lifted 46 of 50 objects, retained a median 94% of the certified grasp margin, and solved the program in 0.09 ms. EquiDexFlow generates SE(3)-equivariant contacts and retargets them to a physical LEAP Hand through inverse kinematics.

My resume, research statement, and curriculum vitae are attached. ParcelStow code and data and models are available online. You can reach me at **enwerem@umd.edu** or **301-416-9430**.

Sincerely,

Clinton Enwerem
October 14, 2026

Clinton Enwerem

Research Statement

enwerem@umd.edu | 301-416-9430 | clintonenwerem.com | GitHub: @coenwerem | LinkedIn | Scholar

Learning and Feedback Control for Dexterous Manipulation

My research concerns how robots should select and execute manipulation actions when contact properties are uncertain and learned behavior changes with execution conditions. I develop grasp representations, state estimators, feedback controllers, and policy evaluations, and implement the software needed to study them in simulation and on physical arms and dexterous hands. I aim to develop manipulation systems that use perception and contact feedback to decide when to continue an action, revise a grasp, or seek additional information.

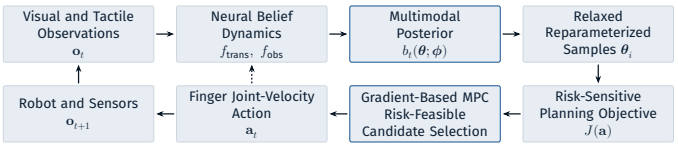


Figure 1. VNB-Grasp: Belief and Feedback Control
Visual/tactile observations update a multimodal belief for risk-sensitive action selection.

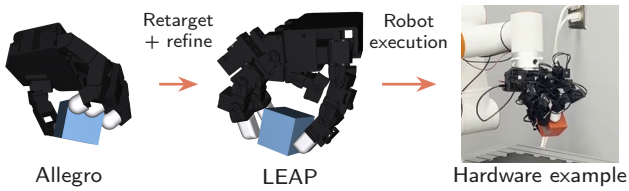


Figure 2. EquiDexFlow: Contact-Based Retargeting
Predicted contacts support retargeting from Allegro to LEAP and hardware execution.

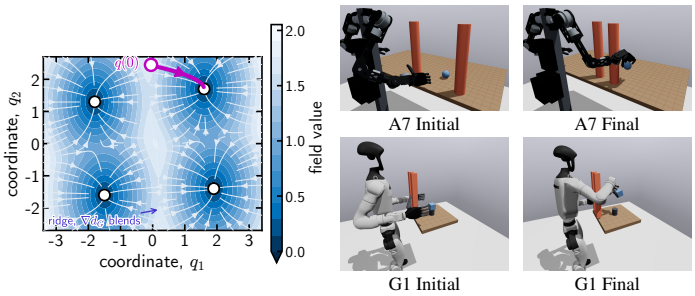


Figure 3. GDF: Field-Guided Grasp Execution
Smooth configuration-space field with initial/final A7 and G1 grasp-execution states in simulation.

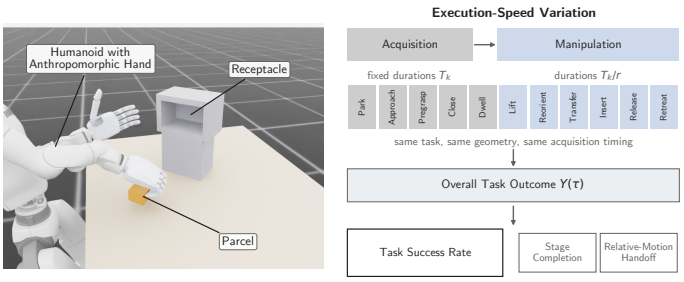


Figure 4. ParcelStow: Controlled Timing Evaluation
Acquisition timing stays fixed while manipulation durations vary; outcomes and task stages are evaluated separately.

Research Contributions

Physics-based reward design and completed reinforcement learning. I trained PPO in 1,024 parallel Isaac Lab environments using grasp-distance-field rewards for a Unitree G1 with a dexterous hand. The policy reached the pregrasp pose in 64/64 evaluation episodes versus 0/64 with palm-distance rewards. The result measures pregrasp reaching, rather than grasp-and-lift completion. My published CDC 2025 work [7] develops CVaR-regularized distributional RL with quantile regression and a contraction analysis. Across 100 MuJoCo reach–avoid episodes, the method reached 448 goals versus 329 and 239 for risk-neutral baselines, with similar constraint-violation costs.

Representing contact uncertainty for grasp selection and control. Object pose, friction, and contact behavior cannot generally be determined from a single observation. My IROS 2026 work, VNB-Grasp [1] (Fig. 1), represents pose and latent contact parameters with an eight-component Gaussian mixture. Relaxed sampling supports pathwise gradients of a smooth conditional value-at-risk (CVaR) objective for action optimization. The representation retains separated hypotheses instead of collapsing uncertainty into a single Gaussian. I evaluated the method across 72 MuJoCo conditions per method, varying friction regime, risk level, randomization, and object. In physical experiments across three objects and four planar pose offsets, VNB-MPC and Gaussian MPC each succeeded in 12 of 12 trials; median episode time was 11.5 s for VNB-MPC and 14.2 s for Gaussian MPC. The experiments distinguish the effects of the belief and controller from the hardware task's success rate.

My complementary work on FIRMGrasp [2] studies grasp selection under friction uncertainty through a CVaR wrench-space margin and a probabilistic force-closure certificate. This formulation makes friction sensitivity part of grasp evaluation. FIRMGrasp and VNB-Grasp examine uncertainty before execution, when selecting a grasp, and during execution, when observations can change the estimated contact conditions.

Belief-informed control under parameter uncertainty. In Belief-Informed Hybrid Control [6], I developed a dual-control algorithm whose two-step lookahead updates the parameter belief from predicted observations before evaluating the next action. An expected-decrease constraint and a bound on cumulative slack permit informative deviations from the target set. Under the stated prediction and

feasibility assumptions, the method guarantees almost-sure target-set convergence at decision times. Simulated bimanual assembly experiments examine how contact feedback changes action admissibility under friction uncertainty.

Feedback execution of feasible grasps. In Grasp Distance Fields [3] (Fig. 3), I use a configuration-space distance field over feasible pregrasp configurations to guide feedback grasp execution. A control-barrier-function/control-Lyapunov-function quadratic program combines progress toward the grasp with collision and workspace constraints. In simulated grasp-and-lift experiments, the controller lifted 46 of 50 objects, correctly rejecting the remaining four grasps as insufficiently stable. The median QP solve time was 0.09 ms. Drake experiments additionally examine reactive execution with changing obstacles and multiple robot models. These simulation experiments examine execution across arm–hand and humanoid models.

Controlled evaluation of imitation policies. I built ParcelStow [4] (Fig. 4) to investigate whether nominal imitation performance persists when task execution timing changes. In Isaac Sim and Isaac Lab, I authored manipulation environments and generated 970K expert states from 297 demonstrations, then trained ACT, Diffusion Policy, and DAgger. The insertion benchmark compares policies with a scripted expert under matched execution speeds and initial conditions. At twice nominal speed, ACT succeeded in 53% of trials compared with 84% for the expert. Most ACT failures occurred during insertion. Stage-level evaluation therefore identifies a precision-manipulation limitation that aggregate nominal success does not reveal. The benchmark supplies a basis for testing whether additional demonstrations, different policy inputs, or feedback corrections address the observed failures.

Learned grasp generation and experimental software. In EquiDexFlow [5] (Fig. 2), I trained an SE(3)-equivariant flow-matching model on 8,100 grasps across 81 objects. I retargeted generated contacts through inverse kinematics to a physical LEAP Hand and evaluated open-loop pick-and-hold execution. This work connects learned grasp generation to embodiment-specific kinematics and physical testing. I also implement ROS 2 hand interfaces, tactile acquisition, VR teleoperation, and motion-planning software. My RealHand L6 software includes a C++ SocketCAN driver, joint and tactile interfaces, contact-gated control, and MoveIt 2 configuration. I verified communication, feedback, and trajectory execution on the physical hand. These capabilities support repeatable data collection and experiments in which algorithmic failures can be distinguished from sensing or execution faults.

Future Research

I would build on these results by studying how evolving pose and contact estimates should change manipulation actions, when failure analysis identifies a need for additional demonstrations, and how learned grasp proposals can be combined with constrained feedback execution. Controlled changes in sensing, contact conditions, and execution timing would separate errors in perception, action selection, and control. Repeatable comparisons with unchanged baselines would test whether each intervention improves task completion while preserving the applicable safety constraints.

Selected References

[1] C. Enwerem et al. *Variational Neural Belief Parameterizations for Robust Dexterous Grasping under Multimodal Uncertainty*. IROS 2026. arXiv:2604.25897.
[2] C. Enwerem et al. *FIRMGrasp: A Friction-Informed Risk Margin for Robust Grasp Synthesis*. Preprint, 2026. arXiv:2607.25049.
[3] C. Enwerem et al. *Grasp Execution Without a Planner: Configuration-Space Grasp Distance Fields with Certified Safety & Guaranteed Quality*. Preprint, 2026. arXiv:2608.00600.
[4] C. Enwerem et al. *Does Imitation Learning Preserve Temporal Robustness in Dexterous Manipulation? An Expert–Learner Comparison Across Task Execution Speeds*. Preprint, 2026. arXiv:2609.01453, project page.
[5] C. Enwerem et al. *EquiDexFlow: Contact-Grounded SE(3)-Equivariant Dexterous Grasp Generative Flows*. Preprint, 2026. arXiv:2606.12728.
[6] C. Enwerem, S. Kemal, J. S. Baras, and C. Belta. *Belief-Informed Hybrid Control with Almost-Sure Target-Set Convergence*. Preprint, 2026. arXiv:2610.07674.
[7] C. Enwerem, A. G. Puranic, J. S. Baras, and C. Belta. *Safety-Aware Reinforcement Learning for Control via Risk-Sensitive Action-Value Iteration and Quantile Regression*. CDC, 2025. doi:10.1109/CDC57313.2025.11312575; arXiv:2506.06954.

Clinton Enwerem

Academic CV

enwerem@umd.edu | 301-416-9430 | clintonenwerem.com | GitHub: @coenwerem | LinkedIn | Scholar

Research Interests

Dexterous Manipulation, Robot Learning, Motion Planning, Risk-Sensitive Control, Safety-Critical Control, and State Estimation under Multimodal Uncertainty.

Education

Ph.D., Electrical & Computer Engineering
University of Maryland, College Park | MD, USA | 2021–Fall 2026
Advisors: Professor John S. Baras and Professor Calin Belta. **Admitted to Candidacy**: June 2025.
Relevant Coursework: Decision-Making for Robotics, Network Control Systems, Decision Making Under Uncertainty, Random Processes, Advanced Digital Signal Processing, Nonlinear Control Systems, Optimal Control, System Theory, Convex Optimization, Formal Methods for Control & Dynamical Systems, Multimodal Foundational Models.

B.Eng., Electrical Engineering
University of Nigeria, Nsukka | Enugu, Nigeria | 2014–2018
Graduated with Highest Honors. *Emphasis*: Control Theory. *Completed coursework in engineering numerical analysis, control systems, electrical machines and drives, computer architecture, electronics, and communication systems*.

Publications

Preprints / Articles In Review

[P5] **Clinton Enwerem**, Saleh Kemal, John S. Baras, and Calin Belta. Belief-Informed Hybrid Control with Almost-Sure Target-Set Convergence, October 2026. Preprint. *2026*

[P4] **Clinton Enwerem**, John S. Baras, and Calin Belta. Does Imitation Learning Preserve Temporal Robustness in Dexterous Manipulation? An Expert–Learner Comparison Across Task Execution Speeds. arXiv preprint, September 2026. arXiv:2609.01453, project page Code Data and Models. *2026*

[P3] **Clinton Enwerem**, John S. Baras, and Calin Belta. Grasp Execution Without a Planner: Configuration-Space Grasp Distance Fields with Certified Safety & Guaranteed Quality, August 2026. arXiV Link Project Page. *2026*

[P2] **Clinton Enwerem**, John S. Baras, and Calin Belta. FIRMGrasp: A Friction-Informed Risk Margin for Robust Grasp Synthesis, July 2026. arXiV Link. *2026*

[P1] **Clinton Enwerem**, John S. Baras, and Calin Belta. EquiDexFlow: Contact-Grounded SE(3)-Equivariant Dexterous Grasp Generative Flows, June 2026. arXiV Link. *2026*

Refereed Conference Publications

[C6] **Clinton Enwerem**, John S. Baras, and Calin Belta. Risk-Constrained Belief-Space Optimization for Safe Control under Latent Uncertainty, December 2026. *To appear in the Proceedings of the 65th IEEE Conference on Decision and Control (CDC).* arXiV Link. *2026*

[C5] **Clinton Enwerem**, Shreya Kalyanaraman, John S. Baras, and Calin Belta. Variational Neural Belief Parameterizations for Robust Dexterous Grasping under Multimodal Uncertainty, 2026. *Proceedings of the 2026 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS).* arXiV Link. *2026*

[C4] **Clinton Enwerem**, Aniruddh G. Puranic, John S. Baras, and Calin Belta. Safety-Aware Reinforcement Learning for Control via Risk-Sensitive Action-Value Iteration and Quantile Regression. *2025 IEEE 64th Conference on Decision and Control (CDC)*, pages 4890–4895, December 2025. arXiV Link. *2025*

[C3] **Clinton Enwerem**, Erfaun Noorani, John S. Baras, and Brian M. Sadler. Robust Stochastic Shortest-Path Planning via Risk-Sensitive Incremental Sampling. *2024 IEEE 63rd Conference on Decision and Control (CDC)*, pages 1087–1094, December 2024. arXiV Link. *2024*

[C2] **Clinton Enwerem** and John S. Baras. Safe Collective Control under Noisy Inputs and Competing Constraints via Non-Smooth Barrier Functions. *2024 European Control Conference (ECC)*, pages 3762–3768, June 2024. *2024*

[C1] **Clinton Enwerem** and John S. Baras. Consensus-Based Leader-Follower Formation Tracking for Control-Affine Nonlinear Multiagent Systems. *9th International Conference on Control, Decision and Information Technologies (CoDIT)*, July 2023. *2023*

Refereed Journal Articles

[J1] **Clinton Enwerem** and John S. Baras. Formation Tracking for a Class of Uncertain Multiagent Systems: A Distributed Kalman Filtering Approach. *IEEE Control Systems Letters*, volume 8, 2024. *2024*

Research Experience

Graduate Research Assistant
Institute for Systems Research, University of Maryland | College Park, MD | 8/2021–Present
Research Topics: *Dexterous Manipulation, Robot Learning, Motion Planning, State Estimation, and Feedback Control*

- Developed risk-sensitive distributional RL (**CDC 2025**): **448 goals** versus **329 and 239** across **100 MuJoCo reach–avoid episodes**, with similar constraint-violation costs.
- Trained PPO with grasp-distance-field rewards in **1,024 parallel Isaac Lab environments**; a Unitree G1 with a dexterous hand reached the pregrasp pose in **64/64** evaluation episodes versus **0/64** with palm-distance rewards.
- Develop grasp-synthesis, state-estimation, motion-planning, and feedback-control methods for dexterous manipulation under friction, pose, and contact uncertainty.
- Derived FIRMGrasp, a CVaR wrench-space margin for grasp selection under friction uncertainty, and implemented Grasp Distance Field control for feedback execution in simulated arm–hand systems.
- Built ParcelStow in Isaac Lab and trained DAgger, Diffusion Policy, and ACT on 970K expert states from 297 demonstrations. At twice nominal execution speed, expert success was 84% and ACT success was 53%.
- Implement Python and C++ software for articulated robot simulation, sampling-based planning, collision checking, constrained optimization, and manipulation control on fixed-base arms and a 21-DOF fixed-pelvis humanoid model.
- Released ROS 2 manipulation software, including realhand_l6_ros2 for tactile dexterous-hand control over SocketCAN and kinematic_planner_ros2 for sampling-based manipulator planning with FCL collision checking.
- Developed VR teleoperation software for an xArm7-RealHand-L6 robotic manipulator with Cartesian velocity control, relative-joint-position hand control, grasp primitives, thumb-abduction control, and hand-state reset; integrated MoveIt Servo with Dimensional OS (DimOS) for collision detection, arm speed modulation, and velocity damping near kinematic boundaries.
- Published first-author work at CoDIT, ECC, CDC, and IEEE L-CSS, with additional papers to appear at IROS 2026 and CDC 2026 and preprints on manipulation and robot learning.

Research Assistant
Electrical Engineering Department, University of Nigeria | Enugu, Nigeria | 9/2018–3/2021
Research Topics: *Robust Control, Observer-Based Compensator Design, Feedback Linearization*

- Designed and implemented robust nonlinear control, active disturbance rejection, feedback linearization, and observer-based compensation for hardware motor control, released as open-source software, ADRC-Motor-Control.
- Co-authored the resulting peer-reviewed journal paper.

Undergraduate Research Assistant
EE Department, University of Nigeria | Enugu, Nigeria | 8/2017–10/2017
Research Topics: *Feedback Control, Time-Delayed Systems, System Identification*

- Designed feedback controllers for first-order-plus-dead-time processes with dead-time compensation and closed-loop system identification, with the implementation and write-up released as foptd-pade.

Professional Experience

Summer Research Assistant
Institute for Systems Research, University of Maryland | College Park, MD | Jun.–Aug. 2023
Supervisor: *Professor John S. Baras*

- Formulated chance-constrained control for multi-agent systems with competing safety constraints.
- Derived Boolean compositions of control barrier functions and evaluated the controller in simulation; the resulting work was published at the 2024 European Control Conference.

Research Intern
MATRIX Lab, USM at Southern Maryland | California, MD | Jun.–Aug. 2022
Supervisor: *Dr. Danilo Romero*

- Conducted system-identification experiments for a Crazyflie quadrotor model.
- Implemented a Lagrangian optimal-control method in ROS-compatible Python for formation tracking with 10 Crazyflie quadrotors under localization uncertainty.

Robotics Engineer
Kognitive Robotics | Lagos, Nigeria | 3/2020–2/2021
Kognitive Robotics is a robotics startup developing turnkey robotics education platforms.

- Designed power and sensor–actuator interface electronics in Altium, assembled the mobile robot, and integrated its sensing and actuation hardware.
- Implemented the ROS software stack and tested hardware–software integration on the physical robot.

Robotics Trainee
Robotics & Artificial Intelligence Nigeria (RAIN) | Ibadan, Nigeria | 1/2020–2/2021

- Built robotics and IoT prototypes spanning mechanical design, sensor fusion, control firmware, embedded software, and product testing.

- Built ROS-based visual-SLAM, state-estimation, and flight-control software for a low-cost UAV on embedded hardware.

Teaching Experience

Graduate Teaching Assistant
ECE Department, University of Maryland | College Park, MD | 9/2025–12/2025
Course: *ENEE467: Robotics Project Laboratory* (with Prof. Calin Belta)
- Mentored 50+ undergraduate students in kinematics, 3D perception, motion planning, and ROS 2 control.
- Built reproducible Docker-based labs and testing pipelines.
- Led weekly lab sessions and provided technical feedback on documentation and debugging.

Graduate Teaching Fellow
ECE Department, University of Maryland | College Park, MD | 2/2025–5/2025
Course: *ENEE661: Nonlinear Control Systems* (with Prof. John S. Baras)
- Led problem-solving sessions on nonlinear analysis and control, covering phase-plane and bifurcation analysis, and stability concepts including Lyapunov, input-to-state, L_2, and L_∞ stability.

Technical Skills

Robotics: ROS/ROS2, ros2_control, MoveIt 2, MoveIt Servo, VR teleoperation, DimOS, MuJoCo, Drake, Isaac Sim, Isaac Lab, Pinocchio, RViz, URDF/SRDF/MJCF, SocketCAN; **Motion Planning:** RRT*, Informed RRT*, Sampling-Based Planning, FCL Collision Checking, OMPL; **Control & Safe Learning:** Safe & Constrained Reinforcement Learning, Control Barrier Functions, Safety Filters, Constrained MDPs, Model Predictive Control (MPC), QP-based Control, Belief-Space Planning, Lyapunov Stability; **Robot Learning & Generative Models:** Imitation Learning & Behavior Cloning, ACT (Action Chunking with Transformers), DAgger, Diffusion Policies (DP3), Flow-Matching Generative Models, Reinforcement Learning (PPO); **ML / Perception:** PyTorch, TensorFlow, OpenCV; **Testing & Validation:** mock hardware, virtual CAN, launch_testing, gtest/pytest, CI; **Programming:** Python, C++, MATLAB, Bash, LaTeX; **Optimization:** CasADi, CVXPY, qpsolvers, IPOPT, Gurobi, Mosek, Pyomo; **RL Sandboxes:** OpenAI Gym, Safety Gymnasium, PyBullet; **Dev Tools:** git, GitHub, GitLab, Docker; **Robots & EoATs:** Fairino FR3, xArm7, UR3e, UR5, Robotiq Hand-E, Unitree G1 (simulation), LEAP Hand v1, RealHand L6, RealHand A7 (Bimanual Arm-Hand System).

Honors & Awards

2024: IEEE CSS Student Travel and Workshop Support: Conference travel award to attend CDC'24.
2022: '22–'23 Microsoft Diversity in Robotics & Autonomy PhD Fellowship, MRC & Microsoft.
2022: ROSCon Diversity Scholarship: Travel grant to attend ROSCon 2022 in Kyoto, Japan.
2021: Finalist, Engineers' League, Pan-African Robotics Competition, Rwanda.
2021: CIT Dean's Fellowship, Carnegie Mellon University, Africa Campus, Kigali, Rwanda.
2021: Dean's Fellowship, University of Maryland, College Park, MD, United States.
2020: Scholar, Stanford Exposure to Research & Graduate Education, Stanford University, CA, USA.
2020: EducationUSA Opportunity Funds Program Scholarship, U.S. Consulate General, Lagos, Nigeria.
2020: Sole Recipient (Nationwide), Door Foundation Leadlight Scholarship, RAIN.
2016–2018: Agbami Science & Technology Scholarship, Chevron: Merit-based undergraduate scholarship.
2015–2018: MTN Foundation Scholarship: Nationwide merit-based undergraduate scholarship.

Professional Training & Development

Certificate in Robot Development & Automation
Robotics & Artificial Intelligence Nigeria (RAIN) | 1/2020–2/2021
Completed graduate-level coursework and projects in robotics, control theory, ML, and IoT.

Open Courseware
Summer 2022, 2023, 2025; Fall 2025
Bayesian Statistics (*UCSD, Coursera*), Autonomous Navigation for Flying Robots (*TUM, edX*), Principles of Robot Autonomy I & II (*Stanford*), Planning & Learning in Robotics (*UCSD*), Robot Learning (*USC*).

Professional Service, Outreach, & Mentoring

2023–2026: *Peer Reviewer*: Heliyon, MED'23, ECC'24, ACC'25, L4DC'25, CDC'25, CDC'26, ICRA'27, IEEE Open Journal of Control Systems, Elsevier (Robotics & Autonomous Systems).
2021–Present: *Member*, Black in Robotics (BiR): the first organization of its kind, advancing Black representation across academic and industry robotics.
2021–2022: *Mentor*, EducationUSA & iScholar Initiative: Guided STEM graduates from underrepresented backgrounds to secure fully-funded graduate admissions at Stanford, Purdue, and Penn State.